Stable Quadratic Pairs over Unique-Base Semirings: Hessenberg Collapse, Supertropical Blocks, and a Wall Obstruction

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Abstract

We construct a stable theory of finite free quadratic pairs over semirings whose free modules have unique bases. Such a pair is a finite weighted graph: vertices carry the quadratic and companion-diagonal coefficients, edges carry the off-diagonal companion coefficients, monomial basis changes act by a unit gauge, and orthogonal sum is disjoint union. The isometry monoid is therefore free commutative on connected gauge-orbit types. Regular pairs split into rank-one loops and rank-two edges; basic metabolic stabilization deletes exactly the metabolic edge generators.

For the closed Hessenberg fragment $\mathcal{H}_d = \{\alpha < \omega^{\omega^d}\}$, natural sum and product identify $\mathcal{H}_d$ with $\mathbb{N}[x_0, \ldots, x_{d-1}]$. Its only unit is one and additive cancellation makes the companion unique and balanced. Every regular pair is consequently a sum of copies of one hyperbolic plane, so the regular Witt group is zero. The all-pairs stable monoid remains nontrivial: it is free on all connected weighted graphs except the hyperbolic edge, and has an exact finite decision procedure.

For the standard rational tangible supertropical cover we give the analogous gauge-orbit presentation, the complete rank-two companion ambiguity, and the regular metabolic generators. Coefficientwise supertropicalization is functorial only for based forms. Two bases of the classical hyperbolic plane give respectively a split edge and two nonmetabolic loops; with the canonical balanced companion the latter are not even regular. Thus no proposed stable class here descends to the classical Witt group without further collapse.

Finally, completing thermograph walls by one global tropical zero produces honest max-plus and min-plus coefficient semifields. Each wall pair defines a regular rank-one quadratic pair classified exactly by its wall gap. The hot switch $\{1 \mid -1\}$ is self-negative but has nonzero gap, proving that the resulting stable wall class cannot be additive under disjunctive sum.

1 Introduction

Over a ring, subtraction hides three different operations in the usual Witt construction: recovery of the polar form from the quadratic form, orthogonal cancellation, and passage from a monoid to a group. None is automatic over a semiring. The correct first object is therefore a quadratic pair (q, b), where

$$q(ax) = a^2 q(x), \qquad q(x+y) = q(x) + q(y) + b(x,y),$$

and b is a specified symmetric bilinear companion [5]. Isometries in this paper preserve both entries of the pair.

The unique-base theorem of Izhakian, Knebusch, and Rowen is the organizing input: if the nonzero elements of a semiring are closed under addition and multiplication, every basis of a free module differs from every other by a permutation and multiplication by scalar units [4, Theorem 1.3]. This replaces matrix reduction by finite graph theory. It also makes clear why the regular theory and the singular envelope behave differently.

The main conclusions are as follows.

Theorem A. *Let stable equivalence kill finite free regular pairs carrying a basic Lagrangian, as in Definition 2.4.*

1. *For every $d \geq 0$, all regular quadratic pairs over the Hessenberg fragment $\mathcal{H}_d$ are hyperbolic. Its regular Witt group is zero. The stable monoid of all pairs is the free commutative monoid on connected weighted-graph types other than the hyperbolic plane.*
2. *Over the standard rational tangible supertropical cover, the all-pairs stable group is free abelian on connected coefficient-graph gauge orbits other than the regular metabolic edge orbits. Regular pairs have only loop and edge blocks, and rank-two companion ambiguity is explicit.*
3. *Canonical coefficientwise supertropicalization neither descends from based forms to unbased isometry classes nor preserves regularity under an arbitrary classical change of basis. Rank parity survives; classification of the universal basis-independent quotient beyond parity remains open.*
4. *The max/min thermograph walls do define regular rank-one stable classes over completed piecewise-linear tropical semifields. Their class is the wall gap, but it cannot be additive in the Conway game group.*

The Hessenberg arm is a complete classification and collapse theorem. The supertropical block theorem is also complete for the stated stable quotient; the remaining issue there is the universal quotient forced by arbitrary unbased scalar extension. The wall theorem is a no-go theorem for the natural max/min construction, not a claim that every possible synchronized cross-quotient is impossible.

2 The unique-base stable monoid

Let R be a commutative semiring such that $R \setminus \{0\}$ is closed under addition and multiplication. Let $V = R^n$ with basis $E = (e_1, \dots, e_n)$, and let $P = (V, q, b)$ be a quadratic pair.

Definition 2.1 (Coefficient graph)**.** The coefficient graph $G_E(P)$ has vertex set $\{1, \dots, n\}$. Vertex i is labelled by

$$(a_i, d_i) = (q(e_i), b(e_i, e_i)).$$

For $i < j$, there is an edge ij precisely when $c_{ij} = b(e_i, e_j) \neq 0$, and that edge is labelled by c_{ij}. Only labelings that occur in a quadratic pair are called admissible.

If $u_i \in R^\times$ and $\pi \in S_n$, the monomial basis change $e_i \mapsto u_i e_{\pi(i)}$ acts by

$$a_i \mapsto u_i^2 a_{\pi(i)}, \qquad d_i \mapsto u_i^2 d_{\pi(i)}, \qquad c_{ij} \mapsto u_i u_j c_{\pi(i)\pi(j)}.$$

We call this the unit gauge action.

Proposition 2.2 (Coordinate expansion)**.** *For $x = \sum_i x_i e_i$,*

$$q(x) = \sum_i x_i^2 a_i + \sum_{i<j} x_i x_j c_{ij}.$$

Thus $G_E(P)$, including the diagonal labels d_i, determines the pair.

Proof. Apply the second quadratic-pair identity repeatedly. Bilinearity gives $b(x_ie_i, x_je_j) = x_ix_jc_{ij}$, and homogeneity gives $q(x_ie_i) = x_i^2a_i$. A bilinear form is determined by its matrix on the basis, so the additional labels d_i determine the full companion. □

Theorem 2.3 (Graph presentation)**.** *Let $\mathcal{C}_R$ be the set of connected admissible coefficient graphs modulo the unit gauge action and labelled-graph isomorphism. The monoid of isometry classes of finite free quadratic pairs satisfies*

$$\mathcal{P}(R) \cong \mathbb{N}^{(\mathcal{C}_R)}.$$

In particular, orthogonal cancellation holds.

Proof. The unique-base theorem makes every module automorphism monomial. By Proposition 2.2, such a map is an isometry of pairs exactly when its permutation and units obey the displayed gauge law. An orthogonal decomposition along subsets of the basis is equivalent to the absence of cross edges. Hence the indecomposable orthogonal summands are exactly the connected components of the coefficient graph. Finite labelled graphs have unique connected-component multisets, which gives the displayed free commutative monoid and cancellation. This is the pair-valued specialization of the component and cancellation theory in [4, Theorems 2.6 and 4.9]. □

Write $b^\flat : V \to V^* = \mathrm{Hom}_R(V, R)$ for $b^\flat(x) = b(x, -)$.

Definition 2.4 (Regular, split, and metabolic)**.** A pair is *regular* if $b^\flat$ is an isomorphism. A submodule is *basic* if it is spanned by a subset of the unique basis. A regular pair is *basic metabolic* if it has a basic submodule L such that $q|_L = 0$ and $L = L^\perp$. It is *basic split* if it has complementary basic Lagrangians.

Let $\mathcal{N}_R$ be the monoid of basic metabolic pairs. Define

$$A \sim_{\mathrm{st}} B \quad \Longleftrightarrow \quad \exists N_1, N_2 \in \mathcal{N}_R : \ A \perp N_1 \cong B \perp N_2.$$

The all-pairs stable group is

$$W_{\mathrm{all}}(R) = K_0(\mathcal{P}(R))/\langle \mathcal{N}_R \rangle.$$

The regular Witt group is defined by the same construction inside the regular submonoid.

Allowing different neutral objects on the two sides is essential. Adding the same object to both sides gives no new equivalence because Theorem 2.3 already proves cancellation.

Theorem 2.5 (Regular block theorem)**.** *Every regular quadratic pair over R is an orthogonal sum of blocks of two kinds:*

$$A(a, d) : \quad q(e) = a, \quad b(e, e) = d \in R^\times,$$
$$H(a_1, a_2; c) : \quad b = \begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}, \quad c \in R^\times.$$

A regular pair is basic metabolic exactly when it has no loop blocks and each edge block has $a_1 = 0$ or $a_2 = 0$. It is split exactly when every block is $H(0, 0; c)$.

Proof. If $b^\flat$ is invertible, its matrix sends a basis of V to a basis of V^*. Unique bases make it a generalized permutation matrix. Symmetry makes the underlying permutation an involution. A fixed point gives a loop block; a transposition gives an edge block.

Let L be basic with $L = L^\perp$. A loop basis vector can neither lie in L, because it is not self-orthogonal, nor outside L, because it would then lie in $L^\perp \setminus L$. In an edge block, equality with the orthogonal complement forces L to contain exactly one endpoint. The condition $q|_L = 0$ says that the selected endpoint has zero quadratic label. Complementary Lagrangians exist exactly when both endpoint labels vanish. □

Consequently, whenever admissibility is decidable, the stable invariant is decidable: decompose the graph, canonically compare the finitely many gauge orbits, delete the metabolic connected edge types, and sort the remaining multiset.

3 The Hessenberg arm

Let # and $\otimes$ denote Hessenberg natural sum and product. Their Cantor-normal-form definitions add coefficients and formally multiply monomials, using natural sum on the exponents [2].

Definition 3.1 (Closed finite-CNF fragment)**.** For $d \in \mathbb{N}$, put

$$\mathcal{H}_d = \{\alpha : \alpha < \omega^{\omega^d}\}.$$

It is equipped with # and $\otimes$, not ordinary ordinal addition or multiplication and not Conway nim arithmetic.

Proposition 3.2 (Polynomial model)**.** *There is a semiring isomorphism*

$$(\mathcal{H}_d, \#, \otimes) \cong \mathbb{N}[x_0, \dots, x_{d-1}].$$

Hence addition is cancellative, nonzero elements are closed under both operations, $\mathcal{H}_d^\times = \{1\}$*,* $a\#a = 1$ *has no solution, and* $a\#a = 0$ *implies* $a = 0$*.*

Proof. Every exponent $\beta < \omega^d$ has the unique expansion $\beta = \sum_{k<d} \omega^k n_k$. Send ω^β to $\prod_{k<d} x_k^{n_k}$, and extend by the finite outer Cantor normal form. Natural sum adds like coefficients, while natural product adds exponent vectors and multiplies coefficients. This proves the isomorphism. All listed properties are immediate in a polynomial semiring over $\mathbb{N}$. □

Fix a quadratic pair over $\mathcal{H}_d$. We continue to write addition and multiplication in semiring notation.

Lemma 3.3 (Unique balanced companion)**.** *The companion is uniquely determined by* q*, and*

$$b(x, x) = 2q(x)$$

for every x*.*

Proof. If b and c are companions, cancel $q(x) + q(y)$ from their two instances of the quadratic-pair identity. For balance, compute $q(x + x)$ in two ways:

$$4q(x) = 2q(x) + b(x, x).$$

Additive cancellation gives $b(x, x) = 2q(x)$. □

Thus a rank-two pair has the unique form

$$Q(a, c, d)(x, y) = ax^2 + cxy + dy^2, \qquad b = \begin{pmatrix} 2a & c \\ c & 2d \end{pmatrix}.$$

Proposition 3.4 (Rank two)**.** *The isometry class of* $Q(a, c, d)$ *is the orbit of* (a, c, d) *under* $(a, c, d) \leftrightarrow (d, c, a)$*. It is decomposable exactly when* $c = 0$*, and regular exactly when*

$$(a, c, d) = (0, 1, 0).$$

Proof. The only scalar unit is one, so every rank-two automorphism is either the identity or the coordinate swap. The graph is disconnected exactly when its one possible edge is absent. If the Gram matrix is invertible, unique bases make it a permutation matrix. Its diagonal entries $2a, 2d$ cannot equal one. Thus it must be the transposition matrix, which forces the displayed triple. The converse is direct. □

Write $H = Q(0, 1, 0)$.

Theorem 3.5 (Hessenberg regular collapse)**.** *Every regular finite free quadratic pair over $\mathcal{H}_d$ is isometric to $H^{\perp m}$ for a unique m. Regular, split, and basic metabolic pairs coincide. Consequently*

$$\mathrm{GW}_{\mathrm{reg}}(\mathcal{H}_d) \cong \mathbb{Z}[H], \qquad W_{\mathrm{reg}}(\mathcal{H}_d) = 0.$$

Proof. Theorem 2.5 reduces the issue to its permutation orbits. A loop would have $1 = b(e, e) = 2q(e)$, impossible by Proposition 3.2. Hence the permutation is fixed-point-free. On a transposition the only nonzero Gram entries are the two off-diagonal ones, both equal to the only unit, one. Its diagonal equality $0 = 2q(e)$ forces both quadratic labels to vanish. Every orbit is therefore H. The remaining assertions follow from the block theorem. □

The singular envelope does not collapse. Let $\mathcal{C}_d$ be the set of connected $\mathcal{H}_d$-weighted graph types from Definition 2.1.

Corollary 3.6 (Complete stable invariant)**.**

$$\mathcal{P}(\mathcal{H}_d)/{\sim_{\mathrm{st}}} \cong \mathbb{N}^{(\mathcal{C}_d \setminus \{H\})}, \qquad W_{\mathrm{all}}(\mathcal{H}_d) \cong \mathbb{Z}^{(\mathcal{C}_d \setminus \{H\})}.$$

There are no further relations.

Proof. Theorem 2.3 gives the free monoid on all connected types, and Theorem 3.5 says that the metabolic submonoid is freely generated by the single type H. Quotienting and group-completing deletes exactly that generator. □

For example, $Q(0, 2, 0)$ and H have identical diagonal data, but the former is a connected generator different from H and survives stably. Thus the singular invariant contains genuine off-diagonal information beyond the forced Cantor-normal-form labels. An exact decision procedure normalizes all CNFs, finds connected components, compares their finitely many vertex permutations, discards H, and compares the sorted multisets.

There is a useful warning about definitions. If regularity is omitted and "metabolic" means only that some basic L satisfies $q|_L = 0$ and $L = L^\perp$, then every $Q(0, c, 0)$ with $c \neq 0$ becomes metabolic: take $L = \mathcal{H}_d e_1$. When c is not a unit the polar map is not surjective. Such a definition would erase singular edge weights for an accidental semiring reason, so regularity or perfect cross-pairing is indispensable.

4 The rational supertropical arm

Let $U_{\mathbb{Q}}$ be the standard tangible max-plus supertropical cover

$$\{0\} \sqcup \{t^\gamma : \gamma \in \mathbb{Q}\} \sqcup \{(t^\gamma)^\nu : \gamma \in \mathbb{Q}\}.$$

Unequal values add by dominance; a tie becomes the corresponding ghost. The tangible elements t^γ are exactly the scalar units, and the value group is dense and two-divisible. The min-plus statement is obtained by $\gamma \mapsto -\gamma$. Free modules have unique bases [5, Theorem 0.9].

For a pair, label vertex i by (a_i, d_i) and edge ij by c_{ij}, as in Definition 2.1. Theorem 2.3 gives the following complete presentation.

Theorem 4.1 (Supertropical stable blocks)**.** *The pair monoid over $U_{\mathbb{Q}}$ is free commutative on connected admissible coefficient graphs modulo*

$$(a_i, d_i, c_{ij}) \mapsto (u_i^2 a_{\pi(i)}, u_i^2 d_{\pi(i)}, u_i u_j c_{\pi(i)\pi(j)}).$$

Every regular pair is an orthogonal sum of loop blocks $A(a,d)$ and edge blocks $H(a_1,a_2;c)$ from Theorem 2.5. Its stable group is free abelian on all connected types except the edge types

$$H(0,a;c), \qquad H(a,0;c), \qquad c \in U_{\mathbb{Q}}^{\times},$$

which are exactly the connected basic metabolic types.

This is a generators-and-relations theorem with no hidden relations. It also gives a decision procedure. For each of finitely many permutations, write $u_i = t^{s_i}$. Equality of the value labels becomes a finite rational linear system in the s_i, together with equality of zero patterns and of tangible/ghost layers. Canonical connected-orbit representatives can therefore be compared in every finite rank.

Companion ambiguity is already visible in rank two. Present a functional quadratic form as

$$q = \begin{bmatrix} a_1 & \alpha \\ & a_2 \end{bmatrix}.$$

Proposition 4.2 (Complete rank-two companions)**.** *All companions have matrices*

$$b = \begin{pmatrix} d_1 & c \\ c & d_2 \end{pmatrix}, \qquad d_i \in [0, ea_i],$$

where

$$c \in \begin{cases} \{\alpha\}, & a_1 a_2 = 0, \\ \{\alpha\}, & \alpha^2 >_\nu a_1 a_2, \\ \{z : z^2 \leq_\nu a_1 a_2\}, & a_1 a_2 \neq 0 \text{ and } \alpha^2 \leq_\nu a_1 a_2. \end{cases}$$

The choices of d_1, d_2, c are independent.

Proof. The diagonal intervals are [5, Proposition 6.12]. If one diagonal coefficient vanishes, the off-diagonal companion is rigid by Proposition 7.9 there. Otherwise the dense, square-divisible specialization of Theorems 7.11 and 7.12 gives the remaining two cases. Independence of the matrix entries is Theorem 6.3 of the same source. □

In particular, one functional q can admit companions with different regular block structures. Stable theory over a supertropical semiring must therefore specify pairs, not only functional quadratic forms.

4.1 The scalar-extension obstruction

Let K be a field of characteristic different from two and let $\varphi : K \to U_{\mathbb{Q}}$ be the standard tangible supervaluation with trivial underlying value on $K^{\times}$. Coefficientwise supertropicalization is functorial for an ordered based form and monoidal under orthogonal sum [5, Definition 9.3]. It is not an invariant of the unbased form.

Consider the classical hyperbolic plane

$$q(xe + yf) = xy.$$

In the basis (e, f), its supertropicalization with the canonical balanced companion is the split edge $H(0, 0; 1)$, hence has stable class zero. In the equally valid basis

$$v_1 = e + f, \qquad v_2 = e - f,$$

one has $q(rv_1 + sv_2) = r^2 - s^2$. Equations (9.11)–(9.13) of [5] give

$$D(1, e) \perp D(1, e), \qquad D(a, ea) : q(\epsilon) = a, \quad b(\epsilon, \epsilon) = ea.$$

The ghost e is not a scalar unit, so both loop blocks are nonregular. They are nevertheless connected generators in the all-pairs group and are not metabolic. Therefore

$$[H(0, 0; 1)] = 0, \qquad 2[D(1, e)] \neq 0.$$

Corollary 4.3 (No unbased canonical extension)**.** *Canonical coefficientwise supertropicalization does not descend to unbased isometry classes, even after regular basic-metabolic stabilization. Moreover, the regular category is not closed under changing the classical input basis.*

Remark 9.5 of [5] supplies a distinct choice: tropicalize the classical polar companion itself. Its diagonal is $\varphi(2a_i)$, rather than $e\varphi(a_i)$, and it is still a companion. In the changed basis above this alternative produces two regular nonmetabolic loops; it repairs regularity in this example but not basis independence. This distinction is also the basis-dependence phenomenon studied explicitly in [6, Section 8].

Rank modulo two is a stable homomorphism: every metabolic generator has even rank, and every basis change preserves rank. It is therefore a classical Witt datum surviving either lift. The example proves that the raw stable group is too fine. We do not claim here that parity is the universal final quotient for arbitrary nonregular canonical images; determining whether more classical Witt data survives all basis relations remains the unresolved part of the supervaluation arm.

5 Thermograph walls

Let Γ be the additive group of continuous functions $f : [0, \infty) \to \mathbb{R}$ that are piecewise affine with finitely many rational breakpoints, rational slopes and intercepts, and are eventually constant. Pointwise maximum, minimum, negation, and division by two preserve Γ. Define

$$\mathbb{P}_{\max} = \Gamma \sqcup \{-\infty\}, \qquad f \oplus g = \max(f, g), \quad f \odot g = f + g.$$

This is an entire zerosumfree idempotent semifield. Its finite elements are all units and its square map is surjective. Negation gives the dual min-plus semifield $\mathbb{P}_{\min} = \Gamma \sqcup \{+\infty\}$.

Let $L, R \in \Gamma$. On the free rank-one $\mathbb{P}_{\max}$-module define

$$q_L(x) = x^{\odot 2} \odot L, \qquad b_R(x, y) = x \odot y \odot R.$$

Theorem 5.1 (Wall pair and gap classification)**.** *The pair (q_L, b_R) is quadratic exactly when $R \leq L$ pointwise. It is regular, and*

$$Q(L, R) \cong Q(L', R') \iff L - R = L' - R'.$$

Proof. Homogeneity and bilinearity are immediate. Pointwise,

$$q_L(x \oplus y) = \max(2x + L, 2y + L),$$
$$q_L(x) \oplus q_L(y) \oplus b_R(x, y) = \max(2x + L, 2y + L, x + y + R).$$

If $R \leq L$, the third term is at most the maximum of the first two. Taking $x = y$ equal to the tropical multiplicative unit proves the converse.

The Gram coefficient R is finite and hence a unit, proving regularity. Every rank-one automorphism is multiplication by a finite unit u, which is an isometry precisely when

$$L = L' + 2u, \qquad R = R' + 2u.$$

This preserves the gap. Conversely, equal gaps give such a unit with $u = (L - L')/2$. □

The left and right thermograph walls of a short game satisfy $R_G \leq L_G$ and belong to Γ [7]. Thus they define $Q(G) = Q(L_G, R_G)$, whose isometry class is exactly the gap $D_G = L_G - R_G$. The dual min-plus construction records the same gap after negation.

In the stable group of Definition 2.4, distinct rank-one loop types survive freely: a basic metabolic regular object has only edge blocks by Theorem 2.5. The necessary basepoint reduction is

$$\Omega(G) = [Q(G)] - [Q(0)].$$

Theorem 5.2 (Hot-switch obstruction)**.** *The map Ω is not additive under disjunctive sum. Consequently the paired max/min wall class is not an additive stable game invariant.*

Proof. Take the switch $X = \{1 \mid -1\}$. Its walls and gap are

$$L_X(t) = \max(1 - t, 0), \qquad R_X(t) = -\max(1 - t, 0), \qquad D_X(t) = 2\max(1 - t, 0).$$

The zero game has gap zero. Theorem 5.1 and freeness of the rank-one generators imply that $\Omega(X) \neq 0$ and has infinite order.

But $-X = X$, so $X + X = 0$ in the Conway game group [3]. Additivity would give $2\Omega(X) = \Omega(0) = 0$, a contradiction. Projection of a paired max/min invariant to its max component would be additive, so the same contradiction rules out the pair. A synchronized quotient can evade it only by killing the nonzero gap atom exhibited above. □

There is a complementary freezing obstruction at temperature zero. Any additive invariant into a cancellative target that depends only on the frozen thermograph assigns the same value to $\uparrow$ and $2\uparrow$. The identity $x = 2x$ then cancels to $x = 0$. Thus wall-only additivity kills the entire common infinitesimal wall class even before the hot-switch argument.

6 Formal boundary and implementation boundary

The Lean module `Ogdoad.SemiringQuadratic` [1] checks:

1. uniqueness and diagonal balance of companions over additively cancellative semirings;
2. the no-half-of-one and no-nonzero-self-double properties of MvPolynomial(Fin $d, \mathbb{N}$), the polynomial model for $\mathcal{H}_d$;
3. that a symmetric balanced permutation Gram matrix is a fixed-point-free involution with zero quadratic basis labels; and

4. that an additive invariant into a cancellative monoid vanishes on any observation identified with its double.

The kernel therefore checks the algebraic core of Theorem 3.5 and the cancellation step in the freezing obstruction. The Cantor-normal-form identification, the unique-base theorem, finite graph decomposition, supertropical companion classification, and standard thermograph facts remain proved here or cited external mathematics, not Lean theorems.

Ogdoad's current `Ordinal` backend implements Conway nim arithmetic and ordinary noncommutative Cantor arithmetic. It is not a Hessenberg semiring. Likewise, the current finite-valued piecewise-linear wall type has no global tropical zero. Reusing either type silently would violate the coefficient hypotheses above. A future implementation should add a distinct finite-CNF natural-arithmetic type and the explicit one-point completions $\mathbb{P}_{\max}, \mathbb{P}_{\min}$.

7 Conclusion

Unique bases turn stable quadratic-pair theory into connected weighted graph theory. In the Hessenberg fragments this gives both extremes at once: the regular Witt theory collapses completely, while the singular stable envelope is a large free invariant with a finite decision procedure. Over the rational supertropical cover the same graph theorem gives exact blocks, companions, and metabolic relations, but coefficientwise supervaluation freezes the chosen classical basis and cannot descend without further quotienting.

Thermograph walls do form honest rank-one quadratic pairs after the minimal tropical completion, and their stable class is exactly the wall gap. The self-negative hot switch proves that this geometrically natural class is incompatible with additivity. The remaining live question is now narrower: determine the universal basis-independent quotient of the nonregular supertropical scalar-extension image, and decide whether it retains anything beyond rank parity.

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